

Tales of Favicons and Caches: Persistent Tracking in Modern Browsers

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The privacy threats of online tracking have garnered considerable attention in recent years from researchers and practitioners alike. This has resulted in users becoming more privacy-cautious and browser vendors gradually adopting countermeasures to mitigate certain forms of cookie-based and cookie-less tracking. Nonetheless, the complexity and feature-rich nature of modern browsers often lead to the deployment of seemingly innocuous functionality that can be readily abused by adversaries. In this paper we introduce a novel tracking mechanism that misuses a simple yet ubiquitous browser feature: favicons. In more detail, a website can track users across browsing sessions by storing a tracking identifier as a set of entries in the browser's dedicated favicon cache, where each entry corresponds to a specific subdomain. In subsequent user visits the website can reconstruct the identifier by observing which favicons are requested by the browser while the user is automatically and rapidly redirected through a series of subdomains. More importantly, the caching of favicons in modern browsers exhibits several unique characteristics that render this tracking vector particularly powerful, as it is persistent (not affected by users clearing their browser data), non-destructive (reconstructing the identifier in subsequent visits does not alter the existing combination of cached entries), and even crosses the isolation of the incognito mode. We experimentally evaluate several aspects of our attack, and present a series of optimization techniques that render our attack practical. We find that combining our favicon-based tracking technique with immutable browser-fingerprinting attributes that do not

change over time allows a website to reconstruct a 32-bit tracking identifier in 2 seconds. Furthermore, our attack works in all major browsers that use a favicon cache, including Chrome and Safari. Due to the severity of our attack we propose changes to browsers' favicon caching behavior that can prevent this form of tracking, and have disclosed our findings to browser vendors who are currently exploring appropriate mitigation strategies.

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